Image Processing and 3D Visualization of TSP and IR Thermography Images

Department of Aerospace Engineering, IIT Kanpur

Dhruv Parikh - 2330540

Mentor by: prof. S.Mariappan

Abstract

To get a flow behavior over a surface the Temperature sensitive paint and infrared thermography techniques are used by capturing the temperature-scaled images of the surface using CCD and Infrared camera, these images are processed to get a clear view of the transition region by extracting out image information and by removing excessive noise from the image using image processing and noise reduction techniques. The transition edge of these images is detected and compared with images at different AOA and with different flow speeds. The processed images are visualized by plotting on a 3D surface by converting 2D IR and TSP images to a 3D model of the NACA0015 airfoil.

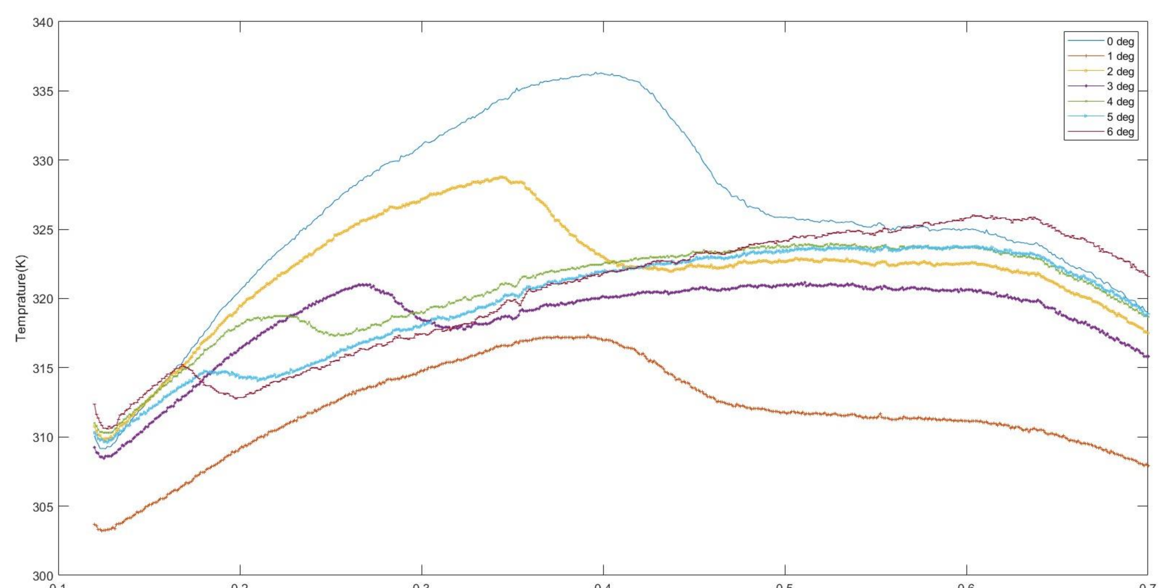
Introduction

To study the flow transition over an airfoil with the help of temperature-sensitive paint(TSP) and infrared imaging. This works on the basis of the temperature variation of the laminar and turbulent regions. This difference is captured on the image of TSP coated airfoil with the help of CCD cameras, from the property of temperature-sensitive paint, it contains luminophores or florescent elements, that emit different intensities of light according to the temperature, it gets excited at a certain wavelength of light and emits light of higher wavelength while returning to ground state but if they are thermally quenched they are at ground state without emitting any light. Another way to measure the temperature difference is by using an infrared camera, it doesn’t require TSP coating, and it can give spital and high-temperature resolution compared to the TSP technique, in IR image value of each individual pixel represents the temperature value at that surface. For the TSP image processing, the reference image is required. Reference images are taken at a constant temperature, and the dark image is taken for the correction of the reference image. When the temperature over the airfoil(NACA0015) is locally constant the wind is turned on and the run image is captured, after this, the image processing is applied to the image pairs in order to obtain relatively divided temperature images. These images are then plotted on the 3D surface of the airfoil for further visualization of transition boundaries and to compare the experimental results with the results obtained from simulation software on the same airfoil model.

Methodology

Methods for Infrared(IR) Image Processing

The IR thermography image captures a clear temperature variation over the surface with a small amount of noise.
The Transition Edge on IR images can be easily detected as the temperature difference between the flow separation bobble, laminar zone, and Turbulent reattachment zone is noticeable, it follows a particular trend of temperature variations over the different angles of attack. The normal transition boundary can be determined by locating the abrupt change in temperature.



Although the same trend doesn’t hold at the forced transition boundary this requires a method of image subtraction in which the image is subtracted by itself with a slight offset. The complete edge is obtained by the addition of both the result images.

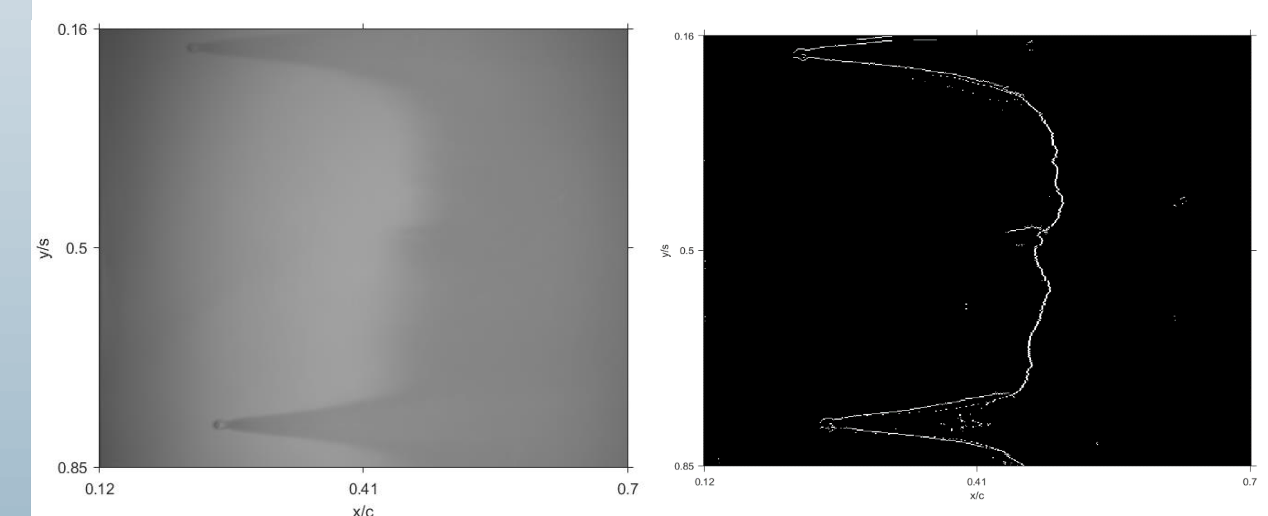
Methods for TSP Image Processing

The TSP image requires different methods to process as it does not follow any of the trends as IR images. The TSP image containing some amount of contrast between the laminar and turbulent zone is filtered with a median filter to concentrate the darker and brighter region, the image is then converted to a binary image at some optimum cutoff value. The binary image is processed with noise reduction filters by reducing noise from darker backgrounds and darker patches from brighter backgrounds but this can expense in the degeneration of transition boundaries and result in a noncontinuous edge, to avoid this the noise reduction filter is applied to a section of an image where it will only apply on the section instead of the entire image. The darker patch in the white region is reduced by applying averaging filter and again converting it to a binary image.

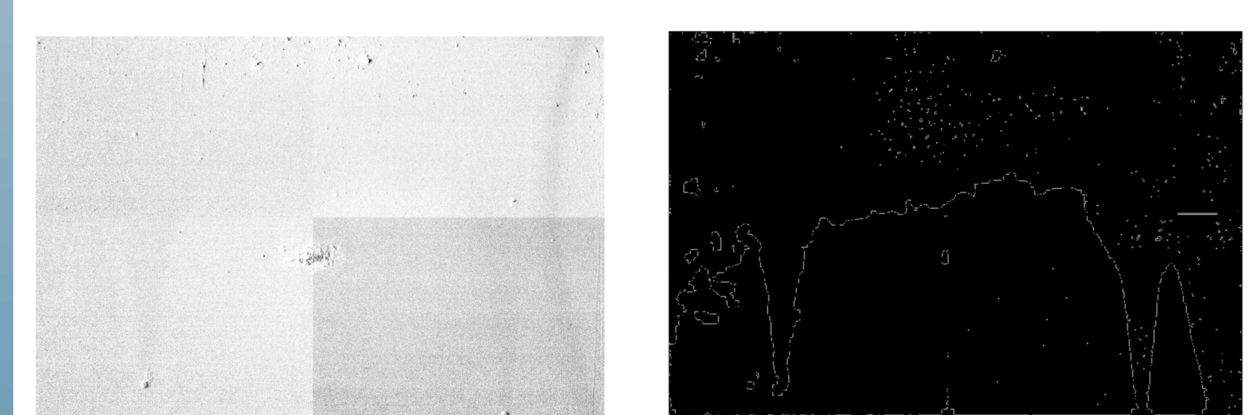
3D surface plotting and visualization

The IR and TSP images are converted to 3D surfaces by using MATLAB surface plotting functions, it takes a mesh grid of the X and Y coordinates and makes the individual surfaces consisting of four points, and each of these individual surfaces represents the Cdata value of an individual pixel of the image, which means the 3D surface of the given image have the same resolution as the image.

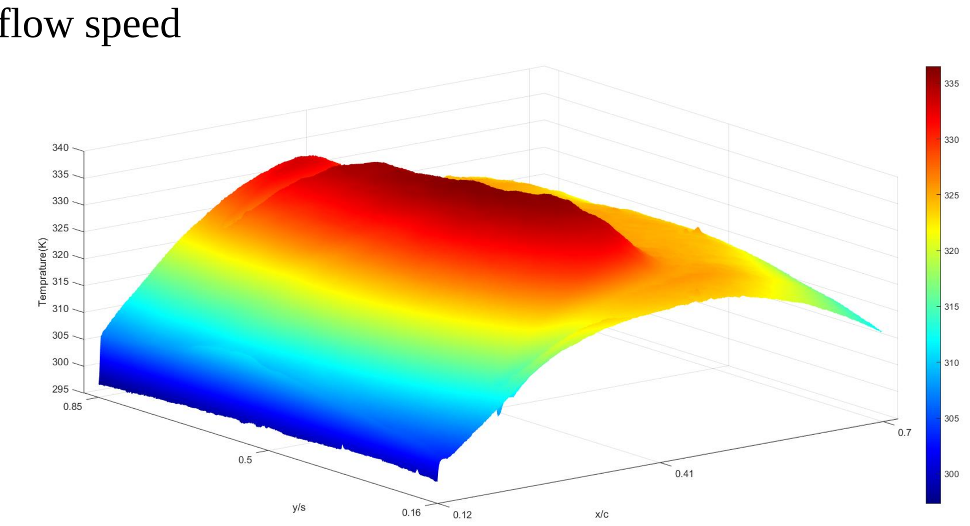
Results and Discussion



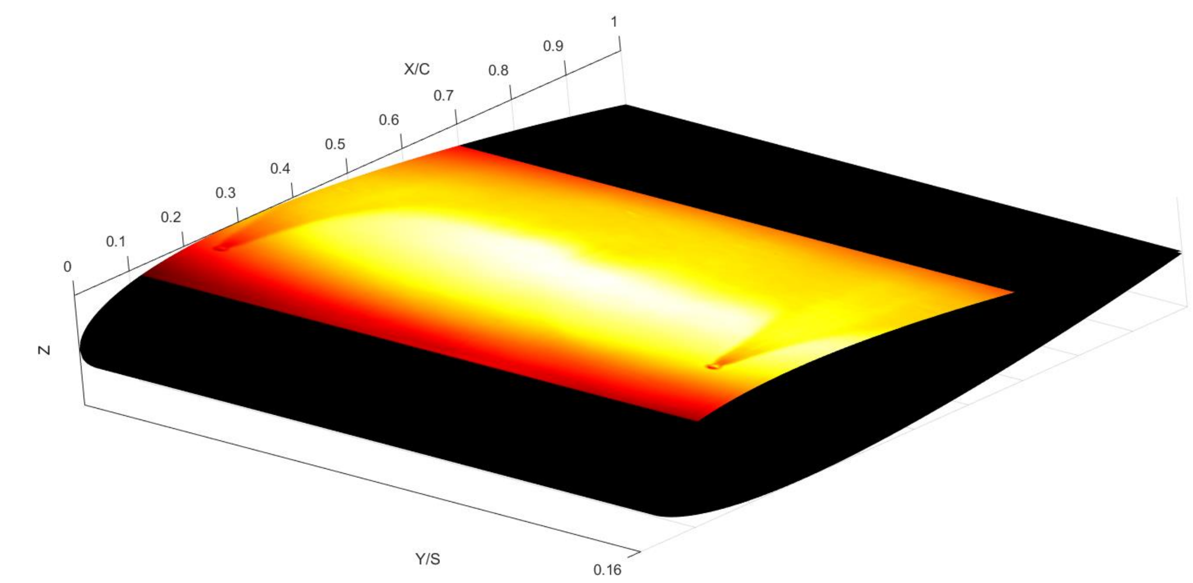
- Results of Edge detection on IR image of NACA0015 model at 0^0 deg angle of attack with 35 m/s flow speed



- Results of Edge detection on Temperature sensitive paint(TSP) image of NACA0015 model at 0^0 deg angle of attack at 35 m/s flow speed



- Temperature variation over the airfoil surface at 0^0 deg angle of attack at 35 m/s flow speed



- Color scale Infrared image at 35m/s flow speed and 0-degree angle of attack, on a 3D model of NACA0015

Conclusion

- IR and TSP image processing was done to extract the laminar to the turbulent transition region.
- Transition Edge was detected for images captured at different angles of attack and at different flow speeds condition.
- The transition edge was compared between different edges at the same angle of attack and different flow speeds and at different AOA with the same flow speed.
- The IR and TSP images were plotted on a 3D model of NACA0015.

References

1. Bestek H, Gruber K, Fasel H (1989) Self-excited unsteadiness of laminar separation bubbles caused by natural transition. The Royal Aeronautical Society, In Proc. Conf. on the Prediction and Exploitation of Separated Flows.
2. Sullivan J., " Temperature and Pressure Sensitive Paint", Lecture Series 2000-2001, Advanced Measurement Techniques, Von Karman Institute for Fluid Mechanics, January 2001.
3. Le Sant Y. and Fontaine J., "Application of Infrared Measurements in the Onera's Wind Tunnels", Wind Tunnels and Wind Tunnel Test Techniques, Cambridge, UK, April 1997.
4. Troyanovsky I., Sadovskii N., Kuzmin M., Mosharov V., Orlov A., Radchenko V. and Phonov S.,"Set of Luminescence Pressure Sensors for Aerospace Research", Sensors and Actuators B, 11 (1993) 201-206.